Four congruence proofs for A396099

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Abstract

This paper gives the defining construction and all completed congruence results for A396099. It includes the full elementary argument for the consecutive-iterate family, so no companion paper is needed to follow the proof. The entry is the parameter $r = 2$.

1 Definition, result, and attribution

For an integer $r \geq 1$, let F_r be the normalized series determined by

$$F_r = x + F_r^{\circ r} F_r^{\circ (r+1)}, \qquad F_r = x + O(x^2). \tag{1}$$

A superscript $\circ k$ means repeated composition; an ordinary power means multiplication. Set $a_n = [x^n]F_2$. The sequence A396099 is this case $r = 2$, with source indices $n \geq 1$ and an additional zero constant coefficient [1]. All congruences below are coefficientwise.

We prove the formula, for every integer k,

$$F_r^{\circ k} \equiv \frac{x}{1-kx} + \begin{cases} \dfrac{rx^4}{(1-x)^3}, & r \text{ even and } k \text{ odd}, \\ 0, & \text{otherwise} \end{cases} \pmod{2r}.$$

The reduced series has exact order $2r$ under composition. Existence, uniqueness, and the meaning of negative iterates are proved below.

Hanna recorded the source sequences. Sundaram proved the $r = 3$ case, including integer iterates, in September 2026 [3]. The odd-parameter argument extends that comparison-series method. The even-parameter argument needs the displayed correction. The source conjecture labels in the retained collection do not establish exclusive priority or worldwide openness.

2 Formal power series and uniqueness

2.1 Definitions

For a commutative ring R with identity, a formal power series is an expression $U(x) = \sum_{n \geq 0} u_n x^n$ with coefficients in R. The notation $R[[x]]$ denotes the set of these series. Equality means equality of every coefficient. No numerical value is assigned to x, so analytic convergence is not involved.

Addition is coefficientwise, and multiplication is defined by

$$[x^n](UV) = \sum_{j=0}^{n} u_j v_{n-j}.$$

When $V(0) = 0$, substitution is defined by $U(V(x)) = \sum_{j\geq 0} u_j V(x)^j$. Every coefficient receives only finitely many contributions. Associativity and the distributive laws then follow coefficient by coefficient from polynomial identities.

A series is *normalized* if its constant coefficient is zero and its coefficient of x is one. Thus it has the form $x + O(x^2)$. The notation $O(x^N)$ denotes a series with zero coefficients in degrees below N. In particular, $U - V = O(x^N)$ says that U and V agree in those degrees. Products satisfy $O(x^a)O(x^b) = O(x^{a+b})$.

For a normalized U, define $U^{\circ 0} = x$ and $U^{\circ(k+1)} = U \circ U^{\circ k}$ for $k \geq 0$. Each iterate is normalized. If $U - V = O(x^N)$, then

$$U^{\circ k} - V^{\circ k} = O(x^N) \qquad (k \geq 0). \tag{2}$$

Indeed, this follows by induction on k: split a difference of compositions into the change in the outer series and the change in the inner series. Substitution by a normalized series preserves the degree of vanishing, and factoring $P^j - Q^j$ shows that a change of order N in the inner argument creates no term of smaller degree.

Reduction of integer coefficients modulo an integer commutes with these operations, because each coefficient is a finite sum of products. A series with constant coefficient one has a multiplicative inverse: if $W(0) = 0$, then

$$(1 + W)^{-1} = \sum_{j\geq 0} (-W)^j.$$

This is well defined coefficientwise, and successive terms cancel upon multiplication. It justifies every fraction used below, whose denominator has constant coefficient one.

2.2 The canonical solution

Lemma 2.1. *For every commutative ring R with identity and every integer $r \geq 1$, there is exactly one normalized series $F \in R[[x]]$ satisfying*

$$F = x + F^{\circ r} F^{\circ(r+1)}.$$

Proof. For normalized U, set $\Phi_r(U) = x + U^{\circ r} U^{\circ(r+1)}$. This series is normalized. Suppose $U - V = O(x^N)$, with $N \geq 2$. By (2), corresponding iterates differ by $O(x^N)$. Each iterate starts with x, so

$$\begin{aligned}\Phi_r(U) - \Phi_r(V) = {} & (U^{\circ r} - V^{\circ r}) U^{\circ(r+1)} \\ & + V^{\circ r}(U^{\circ(r+1)} - V^{\circ(r+1)}) = O(x^{N+1}).\end{aligned}$$

Thus applying Φ_r improves agreement by one degree.

For existence, start with $U_0 = x$ and define $U_{j+1} = \Phi_r(U_j)$. We have $U_1 - U_0 = O(x^2)$; the preceding calculation gives $U_{j+1} - U_j = O(x^{j+2})$ by induction. Hence every coefficient eventually stops changing. Define F by those stabilized coefficients. Each coefficient of $\Phi_r(U)$ depends on only finitely many coefficients of U, so the limit satisfies $F = \Phi_r(F)$.

For uniqueness, two normalized solutions agree below degree two. Whenever they agree below degree N, the displayed calculation and their defining equations force agreement below degree $N + 1$. Induction gives agreement in every degree. □

The proof uses no division, so it applies to residue rings even when the modulus is composite. In particular it constructs the integer series F_r in (1). Its image under any ring homomorphism is the canonical solution over the target ring, by uniqueness. We can therefore identify its reduction modulo $2r$ by finding any normalized solution of the reduced equation.

The first coefficients are useful for checking the indexing:

$$F_r(x) = x + x^2 + (2r+1)x^3 + O(x^4). \tag{3}$$

The product on the right side of (1) starts with x^2, so the second coefficient is one. It follows by induction that $F_r^{\circ j} = x + jx^2 + O(x^3)$. Multiplying the iterates for $j = r$ and $j = r+1$ gives coefficient $2r+1$ in degree three.

3 Comparison series and a substitution identity

Define, for an integer c,

$$T_c(x) = \frac{x}{1-cx}, \qquad T = T_1, \qquad H(x) = \frac{x^4}{(1-x)^3}. \tag{4}$$

These are integer power series. Direct substitution gives

$$T_c \circ T_d = T_{c+d}. \tag{5}$$

Consequently $T^{\circ k} = T_k$ for every nonnegative integer k. The coefficient of x^n in T_k is k^{n-1} for $n \geq 2$, and its coefficient of x is one.

The formal derivative is defined algebraically by $U'(x) = \sum_{n\geq 1} n u_n x^{n-1}$. It satisfies the product and substitution rules, which can be proved first for monomials and then coefficientwise. We need the following rule over a ring in which a particular element has square zero.

Lemma 3.1. *Let $\varepsilon \in R$ satisfy $\varepsilon^2 = 0$, and let $u, v \in xR[[x]]$. For every $S \in R[[x]]$,*

$$S(u + \varepsilon v) = S(u) + \varepsilon v S'(u).$$

Proof. For every nonnegative integer j, the binomial theorem gives $(u+\varepsilon v)^j = u^j + j\varepsilon v u^{j-1}$, with the second term interpreted as zero for $j = 0$. Terms containing two or more factors of ε vanish. Multiply by the coefficient of x^j in S and sum. All substitutions are well defined because their inner arguments have zero constant coefficient. □

There is no division by a factorial in this rule. It therefore works in composite-modulus rings.

Lemma 3.2. *The following identities hold in $\mathbb{Z}[[x]]$:*

$$H(x)T'(T(x)) + H(T(x)) = \frac{2x^4}{(1-2x)^3}, \tag{6}$$

$$(1-x)H(x) - x^2 T(x) - x^3 T'(x) = -\frac{2x^3}{1-x}. \tag{7}$$

Proof. We have $T'(x) = (1-x)^{-2}$, so

$$H(x)T'(T(x)) = \frac{x^4}{(1-x)(1-2x)^2}, \qquad H(T(x)) = \frac{x^4}{(1-x)(1-2x)^3}.$$

Their sum has numerator $x^4((1-2x)+1) = 2x^4(1-x)$ over the common denominator $(1-x)(1-2x)^3$. This proves (6). For the second identity, use denominator $(1-x)^2$ to obtain the numerator

$$x^4 - x^3(1-x) - x^3 = -2x^3(1-x).$$

Cancellation gives (7). □

Finally, expanding three geometric series shows that

$$[x^n]H = \begin{cases} 0, & n < 4, \\ \binom{n-2}{2}, & n \geq 4. \end{cases} \tag{8}$$

The coefficient of x^d in $(1-x)^{-3}$ counts triples of nonnegative integers with sum d. For each first entry i there are $d-i+1$ choices for the other two, giving $(d+1)(d+2)/2 = \binom{d+2}{2}$. This proves (8) without an analytic expansion.

4 The general congruence

Theorem 4.1. *Let $r \geq 1$ and $k \geq 0$ be integers. Then*

$$F_r^{\circ k}(x) \equiv \begin{cases} \dfrac{x}{1-kx}, & r \text{ odd or } k \text{ even}, \\ \dfrac{x}{1-kx} + \dfrac{rx^4}{(1-x)^3}, & r \text{ even and } k \text{ odd}, \end{cases} \pmod{2r}. \tag{9}$$

Proof. First suppose r is odd. Since $r+1$ is even,

$$(1-rx)(1-(r+1)x) = 1-(2r+1)x+r(r+1)x^2 \equiv 1-x \pmod{2r}.$$

Using (5), we obtain

$$x + T^{\circ r}T^{\circ(r+1)} \equiv x + \frac{x^2}{1-x} = T \pmod{2r}.$$

Thus T is the normalized solution of the reduced equation. Lemma 2.1 gives $F_r \equiv T \pmod{2r}$, and composition yields $F_r^{\circ k} \equiv T_k$ for all $k \geq 0$.

Now suppose r is even. Work in the coefficient ring $R = \mathbb{Z}/(2r)\mathbb{Z}$, and set $f = T + rH$. The image of r in this ring has square zero, because r^2 is a multiple of $2r$. Lemma 3.1 and (6) give

$$\begin{aligned} f^{\circ 2} &= T(T+rH) + rH(T+rH) \\ &= T^{\circ 2} + r\big(HT'(T) + H(T)\big) \\ &= T^{\circ 2} + \frac{2rx^4}{(1-2x)^3} = T^{\circ 2} \quad \text{in } R[[x]]. \end{aligned}$$

Because r is even, repeated composition of this identity gives

$$f^{\circ r} = T^{\circ r} = \frac{x}{1-rx} = x + rx^2 \quad \text{in } R[[x]].$$

The last equality again uses $r^2 = 0$ in R. Hence

$$f^{\circ(r+1)} = f(x+rx^2) = f + rx^2 f'.$$

We can now verify the defining equation. After expanding and discarding terms with a factor of r^2, its residual is

$$\begin{aligned} f - \big(x + f^{\circ r} f^{\circ(r+1)}\big) &= (1-x)f - x - rx^2 f - rx^3 f' \\ &= r\big((1-x)H - x^2 T - x^3 T'\big) \\ &= -\frac{2rx^3}{1-x} = 0 \quad \text{in } R[[x]], \end{aligned}$$

where (7) proves the penultimate equality. By uniqueness, the reduction of F_r is f.

For $k = 2j$, the equality $f^{\circ 2} = T^{\circ 2}$ gives $f^{\circ k} = T^{\circ k} = T_k$. For $k = 2j + 1$, it gives

$$f^{\circ k} = f \circ T_{2j} = T_k + rH(T_{2j}).$$

The integer series T_{2j} is congruent to x modulo two. Therefore $H(T_{2j}) - H(x)$ has even integer coefficients, and multiplication by r makes it zero modulo $2r$. Thus $f^{\circ k} = T_k + rH$ in $R[[x]]$, as required. □

Corollary 4.2. *Write $h_n = [x^n]H$ as in (8), and let $\epsilon(r, k)$ equal one if r is even and k is odd, and zero otherwise. For every $r \geq 1$, $k \geq 0$, and $n \geq 2$,*

$$[x^n]F_r^{\circ k} \equiv k^{n-1} + r\epsilon(r, k)h_n \pmod{2r}.$$

The coefficient of degree zero is zero, and that of degree one is one. Every positive-index coefficient of F_r is odd.

Proof. The coefficient formula follows from the geometric series for T_k and (8). At $k = 1$, the correction is absent for odd r and is a multiple of the even integer r for even r. Since 2 divides $2r$, all positive-index coefficients are odd. □

5 Integer iterates and exact composition order

We also prove the integer-iterate extension explicitly, with negative indices defined through the normalized compositional inverse.

Lemma 5.1. *Every normalized series over a commutative ring has a unique normalized two-sided inverse under composition.*

Proof. For $U = x + \sum_{n\geq 2} u_n x^n$, seek $V = x + \sum_{n\geq 2} v_n x^n$ with $U(V) = x$. The coefficient equation in degree n has the form v_n plus an expression involving only earlier coefficients of V. Indeed, in V^j for $j \geq 2$, a coefficient of degree n uses only indices at most $n - j + 1 < n$. Thus every v_n is determined without division, giving a unique right inverse. Apply the construction to V to obtain W with $V \circ W = x$. Associativity gives $U = U \circ (V \circ W) = (U \circ V) \circ W = W$. It follows that $V \circ U = x$ too. □

For an integer $k < 0$, define $F_r^{\circ k}$ by composing the inverse $-k$ times. The usual law of integer powers holds in this composition group. Reducing coefficients commutes with the inverse, because it preserves both inverse identities and the normalized inverse is unique.

Corollary 5.2. *The reduction of F_r modulo $2r$ has exact order $2r$ under composition. Formula (9) remains valid for every integer k, including negative values. In particular,*

$$F_r^{\circ k} \equiv x \pmod{2r} \quad \Longleftrightarrow \quad 2r \mid k.$$

Proof. For nonnegative k, the coefficient of x^2 in the formula of Theorem 4.1 is k modulo $2r$, because H has no terms below degree four. Equality with x therefore forces $2r \mid k$. Conversely, if $2r \mid k$, then k is even, $T_k \equiv x \pmod{2r}$, and the correction term is absent. This proves the order statement.

For any integer k, choose the unique j with $0 \leq j < 2r$ and $j \equiv k \pmod{2r}$. In the composition group modulo $2r$, the order statement gives $F_r^{\circ k} = F_r^{\circ j}$. Also $T_k \equiv T_j \pmod{2r}$, and k and j have the same parity, because $2r$ is even. The formula for j is therefore exactly the required formula for k. The same coefficient-of-x^2 argument proves the final equivalence for all integer k. □

6 All four A396099 assertions

Take $A = F_2$ and put $T = x/(1-x)$, $H = x^4/(1-x)^3$. The general theorem gives

$$\begin{aligned} A &\equiv T + 2H \pmod 4, \\ A^{\circ 2} &\equiv x + 2x^2 \pmod 4. \end{aligned}$$

Every positive-index coefficient of the first expression is odd. Since $[x^n]H = \binom{n-2}{2}$ for $n \geq 4$ and is zero below degree four, the residue of a_n for $n \geq 3$ is three if $n \equiv 0, 1 \pmod 4$ and one otherwise. Indeed, $\binom{n+2}{2} - \binom{n-2}{2} = 4n - 2$ is even, and the values at $n = 2, 3, 4, 5$ are $0, 0, 1, 3$. Thus the residue block is $1, 3, 3, 1$ beginning at degree three. The second displayed identity shows that every coefficient of the second iterate above degree two is divisible by four.

For the remaining assertion work modulo four and write $f = T + 2H$. From $f^{\circ 2} = x + 2x^2$ and the square-zero substitution rule, $f^{\circ 3} = f + 2x^2 f'$. Therefore

$$\begin{aligned} f - x f^{\circ 3} &= (1-x)f - 2x^3 f' \\ &= x + 2\big((1-x)H - x^3 T'\big) \\ &= x + 2\frac{x^4 - x^3}{(1-x)^2} = x + \frac{2x^3}{1-x} \pmod 4. \end{aligned}$$

Every coefficient above degree two in $A - xA^{\circ 3}$ is consequently two modulo four. This proves all four source assertions, including their exact starting indices. The separate source contract also proves them for any normalized integral series satisfying the original equation, by the uniqueness already established.

7 Formal proof and scope

The accompanying entry file is `lean/Entries/A396099.lean`. It states the results with explicit Lean types and supplies proof terms using the complete shared implementation included in the collection. The manuscript is self-contained mathematically; the Lean entry file uses those ordinary library imports rather than duplicating the full foundation.

The entry's named theorem interfaces are listed below. The namespace of each is `FinalCollection.A396099`.

- `source_existence_uniqueness`
- `source_normalized`
- `source_equation`
- `exact_composition_order`
- `a396099_all_four`
- `a396099_odd`
- `a396099_residue`
- `a396099_period`
- `a396099_second_iterate_series`
- `a396099_second_iterate`
- `a396099_difference_series`
- `a396099_difference`

The preserved dependencies also contain all supporting constructions, coefficient identities, map-compatibility results, and inverse proofs. The collection pins Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The verification report records fresh compilation and the transitive axiom audit. Finite numerical checks corroborate the definitions and indexing; the universally quantified Lean theorems establish the all-index claims. No assertion of worldwide novelty, OEIS acceptance, publication, or completed human review follows from these checks. The prose may use elementary identities that the formal proof establishes by a different algebraic route.

References

[1] P. D. Hanna, *A396099*, The On-Line Encyclopedia of Integer Sequences. `https://oeis.org/A396099`. Source recorded in the retained proof collection, 24 October 2026.

[2] D. Goyal, *OEIS A396843 proof repository.* `https://github.com/dvgyl/oeis-a396843-proof`. Reference revision `995e55adcbe06b2a913e24f12ff73a7faf362f06`.

[3] Shiva Priyan Shanmuga Sundaram, *A Proof of the Congruence Conjectures for OEIS A396797*, September 2026. `https://oeis.org/A396797/a396797.pdf`.